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Brain weight across the adult lifespan in great apes

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Reduction in brain weight is a hallmark feature of human aging, which is exacerbated in pathologies, such as Alzheimer's disease and other dementias. However, it remains unclear whether a decrease in brain weight with age reflects a vulnerability unique to the human lineage or represents a consequence of an extension of aging processes shared with other primates. To address this question, we examined age-related changes in brain weight across our closest living relatives, the great apes (chimpanzees, bonobos, gorillas, and orangutans). The sample spanned the full adult age range in each species and included individuals reaching ages comparable to those at which age-related brain changes become apparent in humans. Across species, brain weight did not show significant change throughout adulthood and into advanced age. The observed pattern held even when body weight was included in the analysis in a subsample of chimpanzees. Overall, these results indicate that marked brain weight loss is not a general feature of great ape aging and instead appears largely specific to humans, with implications for understanding susceptibility to late-life brain pathology.

brain | aging | evolution | primates | Alzheimer's disease

Significance

Even in the absence of pathologies, human brains are characterized by age-related changes in structure, which eventually become so pronounced that a decrease in overall brain size is apparent. These changes may underlie uniquely human susceptibility to neurodegenerative diseases late in life. The evolutionary origins of this vulnerability remain poorly understood. We approach this problem by examining age-related changes in brain weight in our closest primate relatives, the great apes (chimpanzees, bonobos, gorillas, and orangutans). We did not observe significant decrease in brain weight with age in these species. This indicates that the reduction in brain mass that is typical of normal human aging is not present in great apes, potentially linked to our species' elevated risk for neurodegenerative conditions.

Brain aging in humans represents an evolutionary conundrum, as a key feature of human life history, namely a long lifespan, can compromise neural systems that support cognitive function with age. Even in the absence of disease, normal human aging is accompanied by cognitive changes (reviewed in 1), including alterations in information processing (2–4) and shifts in brain organization compared to younger individuals. Research has shown that aging in humans involves a decrease in cerebral volume (5, 6) in the gray matter (GM) and white matter (WM; 7, 8), reduction in the size of specific cortical areas and subcortical structures (9–11), and various microscopic changes, such as decreased complexity of cortical pyramidal neuron dendritic arbors and lower density of synaptic spines (12, 13). The extent to which such age-related changes are also present in other primates is an open question, as brain aging in nonhuman primates displays variation among structures that appear species-specific and distinct from patterns observed in humans (14–16). Distinguishing the features associated with aging shared with other primate species from those that are uniquely human represents a crucial step in our understanding of the factors that make certain brain regions (17–20) or classes of neurons (21–23) prone to selective degeneration, leading human brains to be especially vulnerable to pathologies in later life.

Decrease in overall brain size represents one feature that might be unique to aging in humans among primates. Both in vivo MRI studies of brain volume and postmortem autopsy studies of brain weight demonstrate a progressive decline in overall brain size in humans from adulthood into old age. In an extensive review of more than 50 MRI longitudinal studies, Hedman et al. (24) reported a 0.2% per year decrease in overall brain volume from 30 years of age onward, which, after the age of 60 increases to 0.5% of volume loss per year. Focusing specifically on aged individuals (59 y and older), Resnick et al. (25) reported a decrease of 5.4 cm^3 per year in brain volume on average. However, it is important to note potential differences in the rate of brain volume decrease across human populations. For example, it has been reported that the rate of decrease in brain volume among Tsimane, a forager/horticulturalist population in lowland Bolivia, is lower compared to European and US populations (26), highlighting the role of diet, activity, exposure to pathogens, and other environmental factors on brain aging.

Postmortem studies of human brain weight typically report lower values for change with age, although the pattern of the gradual decline remains 2.7 g per year in males and 2.2 g per year in females (27) that increases to 5 g per year after the age of 80 y (28). An often-cited study (29) utilizing a large sample of over 4,000 autopsies reported a gradual decrease in brain weight starting between 45 and 50 years of age. Quantifying the decrease

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as a percentage, the authors noted an 11% decrease in brain weight between 19 and 86 years of age. A similar decrease in brain weight, albeit with an earlier onset (30 y), has been reported from the analysis of 2,080 hospital autopsy records, which included 691 subjects older than 70 y and 39 subjects between the ages of 85 and 96 y (30). In a series of 724 forensic cases, Pakkenberg and Voigt (31) reported a decrease of 100 g in brain weight between 25 and 70 years of age in males. In contrast, based on a study utilizing a sample of 399 rhesus macaques (*Macaca mulatta*), which included individuals toward the later stages of the lifespan, aged 25 and older (32), brain weight appeared to show no change with age.

There are few data on aging-related changes in brain size among humans' closest living primate relatives, the great apes (chimpanzees, bonobos, gorillas, and orangutans). The evolutionary split between humans and the other great apes occurred approximately 6 to 8 million years ago (Mya; (33), which is considerably more recent than between humans and cercopithecoid monkeys, including macaques (30 Mya; 34). Importantly, great apes have substantially shorter lifespans than humans, raising the question of whether age-related brain atrophy is a uniquely human trait linked to our extended postreproductive longevity (35, 36). This comparison is essential for placing human brain aging in evolutionary context. A previous study employing a sample of 76 chimpanzees (*Pan troglodytes*) including individuals in their 40s and 50s, reported a decrease in brain weight of 1.1 g per year (37). However, when males and females were analyzed separately, the decrease in brain weight was not statistically significant. These findings suggest that if brain weight declines with aging in chimpanzees at all, it may be well below the amount reported for humans.

In the present study, we analyzed brain weight changes with age in four great ape taxa utilizing a sample that covers the adult lifespan. The dataset is the largest and most complete of its kind, which allowed us to: 1) revisit the extent of changes in brain weight throughout the lifespan in chimpanzees, by including a significantly larger sample with more geriatric specimens compared to previous studies, and 2) determine whether age-related decline in brain weight is found in other great ape species, namely bonobos (*Pan paniscus*), western lowland gorillas (*Gorilla gorilla*), and orangutans (*Pongo pygmaeus* and *Pongo abelii*).

Results

Age-related changes in brain weight were analyzed in females and males separately in each species due to sexual dimorphism and differences in lifespan where females represented a larger proportion of the oldest individuals (Table 1). The sample included subjects beginning at 4 years of age, which is when total brain size in great apes typically approaches adult-like levels (*Materials and Methods*). The substantial variation in brain weight observed within species is consistent with previously reported variation in great ape endocranial volume, where the largest individuals may be approximately twice the cranial capacity of the smallest adults within the same species (38). To assess the relationship between brain weight and age, we fit least squares linear regressions for each group and report the slope to indicate the rate of change (g/year), coefficient of determination (R^2), *P*-value, and 95% CI for the slope. Slopes that had 95% CIs which included zero were considered to indicate stable brain weight with age. To evaluate potential nonlinear age-related patterns, we first tested quadratic age models against linear alternatives for each species and sex using the Akaike Information Criterion (AIC). In no case did quadratic models improve fit (all $\Delta AIC < 2$), and linear models were therefore retained for parsimony. We additionally conducted objective piecewise regressions allowing a single breakpoint selected by Bayesian Information Criterion (BIC) from the central 80% of the observed age range; however, no breakpoint model was supported ($\Delta BIC \leq 2$). Accordingly, only linear model results are reported below (Table 2).

Table 1. List of species with basic demographic characteristics of the sample used in this study

Species	Number of specimens (age range)
Chimpanzee (*Pan troglodytes*)	
Female	236 (5–70 y)
Male	221 (5–61 y)
Total	457
Bonobo (*Pan paniscus*)	
Female	8 (9–62 y)
Male	8 (4–49 y)
Total	16
Western lowland gorilla (*Gorilla gorilla*)	
Female	36 (17–64 y)
Male	41 (8–61 y)
Total	77
Orangutan (*Pongo pygmaeus*, *Pongo abelii*, *Pongo* hybrid)	
Female	23 (10–61 y)
Male	28 (5–48 y)
Total	51

Chimpanzees. The sample of chimpanzees included 457 individuals (females: n = 236; males: n = 221; age range 5 to 70 y), with 123 individuals above the age of 40 y and 42 over the age of 50. Among females, no significant age-related change in brain weight was detected (slope = −0.238 g/y, 95% CI: −0.640 to 0.164, $P = 0.243$, $R^2 = 0.006$). Similarly, males showed no significant age-related change (slope = 0.011 g/y, 95% CI: −0.530 to 0.551, $P = 0.969$, $R^2 < 0.001$). In both sexes, slopes were near zero, indicating no change in brain weight across the adult lifespan. Because the age at which adult brain size is attained may vary among individuals, we evaluated the sensitivity of the results to the minimum age selected for inclusion by repeating the analyses across a moving age threshold up to 15 years of age, thereby progressively excluding younger individuals that may still have been undergoing developmental maturation. Estimated age-related slopes varied little across thresholds, ranging from −0.23 to −0.47 g/y in females and from +0.01 to −0.30 g/y in males, indicating that exclusion of progressively younger individuals had minimal influence on the inferred relationship between age and brain weight. Effect sizes remained small at all thresholds (maximum $R^2 = 0.021$), demonstrating that the overall conclusions were stable given the specific minimum age cutoff used to define the adult sample.

ANCOVA testing for sex differences in age-related slopes (Age × Sex) revealed no significant interaction ($F_{1,453} = 0.54$, $P = 0.463$), indicating that males and females did not differ in aging patterns (Fig. 1*A* and Table 2). There was a significant main effect of sex on brain weight ($F_{1,453} = 45.89$, $P < 0.001$), consistent with sexual dimorphism, but no evidence of differential aging trajectories.

To evaluate age-related variation in brain weight further by taking body size into account, we conducted additional analyses in a subset of chimpanzees for which body weight recorded at necropsy was also available. Analyses were restricted to individuals aged ≥15 y, corresponding to the approximate attainment of adult body

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Table 2. Summary of least squares linear regression results for brain weight changes with age in adults of each species

Species	Age-related change in brain weight – slope (g/year)	Slope 95% CI (lower)	Slope 95% CI (upper)	R^2	*P*-value
Chimpanzee (*Pan troglodytes*)					
Females	−0.238	−0.640	0.164	0.006	0.243
Males	0.011	−0.530	0.551	<0.001	0.969
Bonobo (*Pan paniscus*)					
Females	0.060	−2.488	2.609	0.001	0.956
Males	−0.885	−3.546	1.777	0.099	0.447
Western lowland gorilla (*Gorilla gorilla*)					
Females	0.632	−0.842	2.105	0.022	0.390
Males	−0.587	−2.391	1.217	0.010	0.514
Orangutan (*Pongo pygmaeus*, *Pongo abelii*, *Pongo* hybrid)					
Females	0.295	−1.226	1.817	0.008	0.690
Males	0.442	−1.949	2.832	0.006	0.707

The table shows the slope, 95% CI, coefficient of determination (R^2), and *P*-values. Significance is set at $\alpha = 0.05$.

weight in chimpanzees, to focus on changes occurring across the adult lifespan rather than during developmental growth. Brain weight was positively associated with body weight in both sexes (females: slope = 0.570 g/kg, $R^2 = 0.061$, $P = 0.003$, n = 147; males: slope = 0.883 g/kg, $R^2 = 0.086$, $P = 0.0007$, n = 131; Fig. 2 *A* and *D*), indicating brain–body scaling in adulthood. However, body weight itself showed no significant relationship with age for females or males (females: slope = 0.009 kg/y, $R^2 < 0.001$, $P = 0.933$; n = 147; males: slope = –0.210 kg/y, $R^2 = 0.019$, $P = 0.117$; n = 131; Fig. 2 *B* and *E*). Therefore, to assess age-related brain changes accounting for body size, we examined residual brain weight (from brain weight ~ body weight) in relation to absolute age. Residual brain weight showed no statistically significant decline with age in either females or males (females: slope = –0.321 g/y, $R^2 = 0.013$,

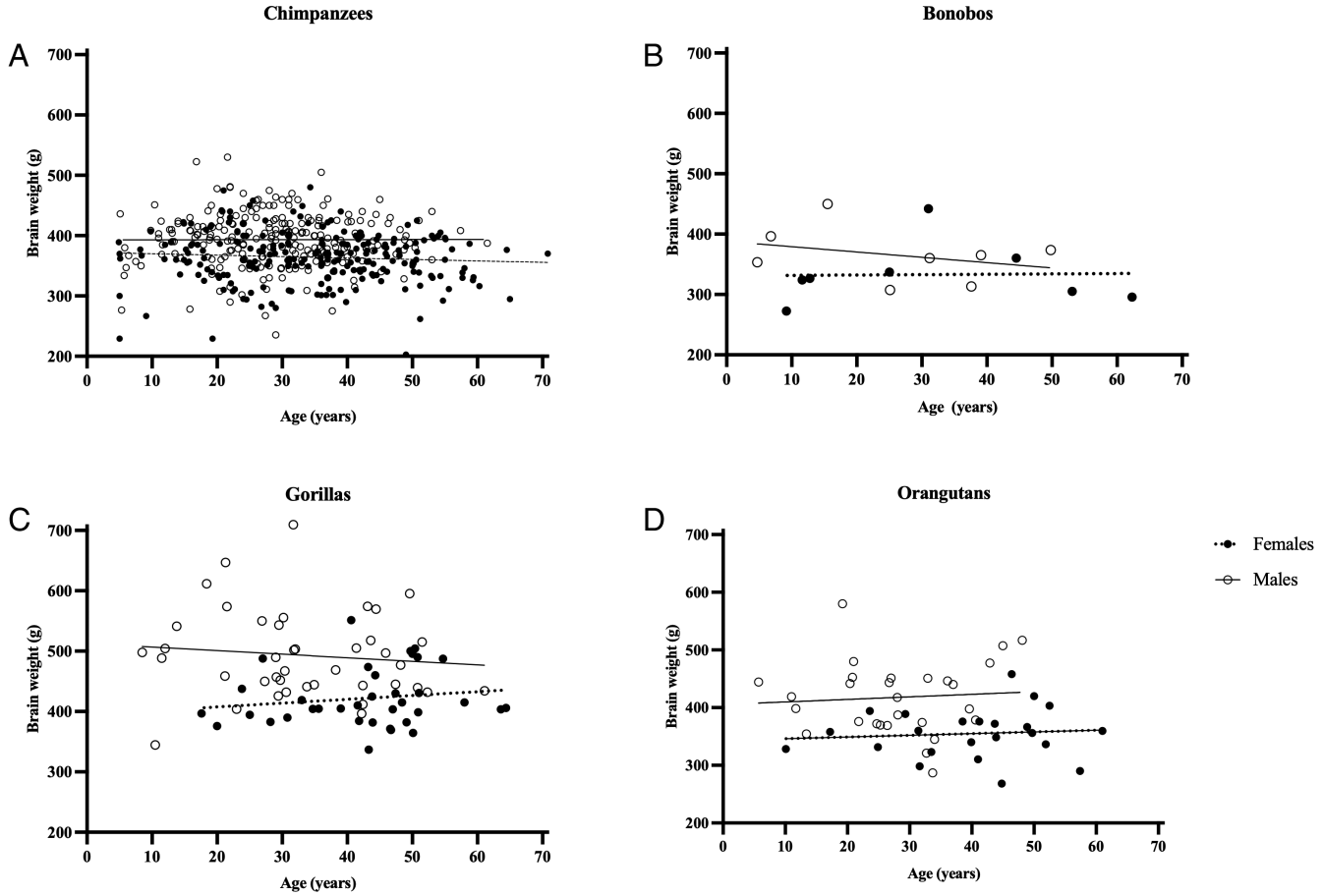


Fig. 1. Least squares regression of brain weight changes with age in adult specimens: chimpanzees (*A*), bonobos (*B*), western lowland gorillas (*C*), and orangutans (*D*). Closed circles represent females; open circles represent males.

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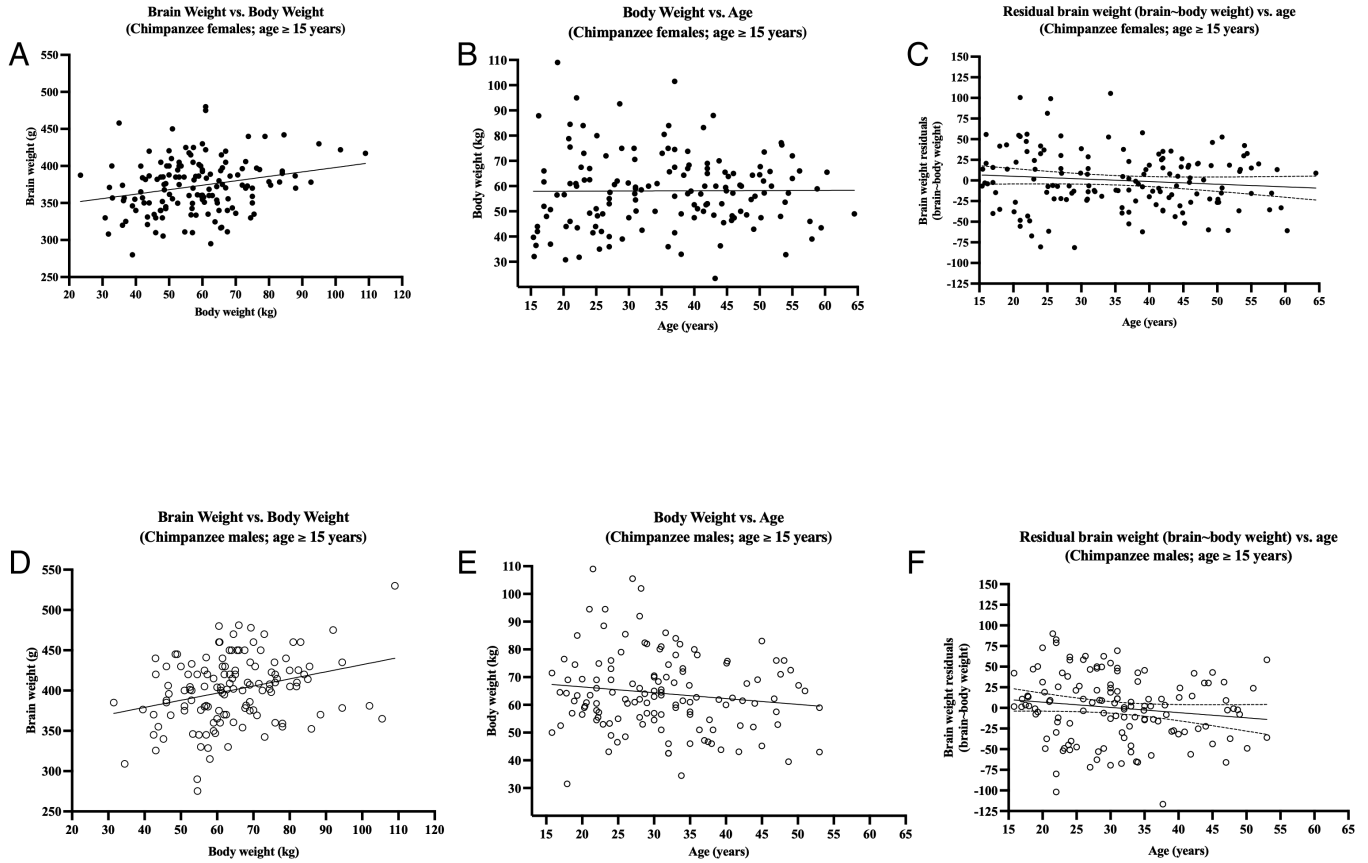


Fig. 2. The relationship between body weight, brain weight, and age in female (*A*–*C*) and male (*D*–*F*) chimpanzees. Closed circles represent females; open circles represent males.

P = 0.176; n = 147; males: slope = –0.633 g/y, R^2 = 0.021, P = 0.102; n = 131; Fig. 2 *C* and *F*).

Bonobos. In a sample of 16 individuals (females: n = 8; males: n = 8; age range 4 to 62 y), there were no significant age-related brain weight changes in bonobos (females: slope = 0.060 g/y, 95% CI: –2.488 to 2.609, P = 0.956, R^2 = 0.001; males: slope = –0.885 g/y, 95% CI: –3.546 to 1.777, P = 0.447, R^2 = 0.099). ANCOVA tests of sex differences in slopes were not performed given limited data across the adult age span (Fig. 1*B* and Table 2).

Gorillas. The sample of gorillas included specimens between 8 and 64 years of age, with 14 individuals older than 50 y. Neither females (n = 36) nor males (n = 41) exhibited a significant overall trend for change in brain weight with age (females: slope = 0.632 g/y, 95% CI: –0.842 to 2.105, P = 0.390, R^2 = 0.022; males: slope = –0.587 g/y; 95% CI: –2.391 to 1.217, P = 0.514, R^2 = 0.010). ANCOVA showed no significant sex differences in slopes ($F_{1,72}$ = 0.97, P = 0.328; Fig. 1*C* and Table 2).

Orangutans. The orangutan sample consisted of 51 individuals (females: n = 23; males: n = 28; age range 5 to 61 y). Neither sex showed a significant overall age-related change in brain weight (females: slope = 0.295 g/y, 95% CI: –1.226 to 1.817, P = 0.690, R^2 = 0.008; males: slope = 0.442 g/y, 95% CI: –1.949 to 2.832, P = 0.707, R^2 = 0.006). ANCOVA indicated no significant sex difference in slopes ($F_{1,47}$ = 0.01, P = 0.916; Fig. 1*D* and Table 2).

Comparing species-level differences across all great apes (n = 601), ANCOVA (including sex as a factor) tested for an Age × Species interaction and found no significant effect ($F_{3,591}$ = 0.45, P = 0.719). There was likewise no significant Age × Sex interaction ($F_{1,591}$ = 0.51, P = 0.474). The main effect of age was not significant ($F_{1,591}$ = 0.98, P = 0.324). Thus, we can conclude that estimated age-related slopes were similar across species and sexes, and the interaction terms provided no evidence that these relationships differed among groups (Table 2).

Discussion

Across the great apes, we found that brain weight does not significantly decrease from the age at which adult brain size is attained into old age. In comparison, humans exhibit marked reductions in brain size beginning between 45 to 50 years of age and accelerate thereafter (27–31). The differences between humans as reported in the literature (27–29) and the great apes analyzed here is unlikely to be driven by variation in the ages represented in the samples, as a sizable number of geriatric specimens over the age of 50 y were included (14 gorillas, 5 orangutans, and 42 chimpanzees). The only species with a relatively small proportion of older individuals was bonobos, where only 2 of 13 adult specimens were over the age of 50 y. Although sample sizes differed substantially among species, particularly for bonobos and orangutans, the results were consistent across all great apes examined. Future studies with larger samples will be important for refining species-specific estimates of age-related change and for detecting subtle effects that may not be apparent in the currently available datasets. Notably, these results are based on individuals with a wide range of causes of death, encompassing older great apes known to exhibit age-related neuropathology such as phosphorylated tau and amyloid β accumulation (39). Therefore, the absence of detectable

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age-related decline in brain weight even within this mixed sample is remarkable, suggesting that overall brain mass is relatively preserved despite the likely presence of such pathology in some individuals. In future work, it will be important to compare trajectories of brain aging between individuals with and without neuropathological changes; however, subdividing the sample in this manner will require substantially larger datasets to achieve sufficient statistical power for such comparisons.

As with any cross-sectional study, our analyses may underestimate age-related change if individuals with relatively smaller brains or accelerated brain aging are less likely to survive into older age classes. Such selective disappearance could attenuate the appearance of age-related trends by causing older individuals to represent a nonrandom subset of the population. However, evaluating this possibility directly requires longitudinal measurements of brain size within individuals. Although longitudinal neuroimaging studies have been instrumental in characterizing human brain aging, comparable datasets are unlikely to become available for great apes because repeated MRI examinations are constrained by ethical and practical limitations. Nevertheless, the absence of detectable age-related decline in brain weight observed here contrasts with both cross-sectional autopsy and longitudinal MRI studies in humans, which demonstrate measurable reductions in brain size with advancing age. Thus, although survivorship effects cannot be excluded, they are unlikely to fully account for the difference currently observed between humans and other great apes.

Our findings indicating a lack of significant age-related brain weight changes in a large sample of chimpanzees differ from a previous report in which males and females were analyzed together (37). However, as female brain weights are generally lower than males, this apparent relationship was likely driven by a higher proportion of females surviving into old age, which is consistent with our data (Fig. 1*A* and Table 2).

Differences in the patterns of aging between humans and other primates are typically attributed to a prolonged life span. In a study examining cortical aging in humans and chimpanzees (15), it was noted that the structures displaying differences—frontal WM, neocortical WM, and the hippocampus—started decreasing in elderly humans older than the oldest chimpanzees included in that analysis. A separate study similarly did not find changes in overall GM and cortical thickness associated with age in chimpanzees (40). Recent comparative neuroimaging analyses further indicate that the spatial pattern of age-related gray matter decline in humans is strongly associated with regions that experienced the greatest evolutionary cortical expansion, particularly in higher-order prefrontal areas, whereas chimpanzees do not show this relationship (18). This finding suggests that brain regions that expanded most during recent human evolution may also be disproportionately vulnerable to aging-related atrophy. The oldest chimpanzees in these studies were in their 40s, that is, past the typical life expectancy of around 30 for chimpanzees in the wild (41) and at the upper range of those reported for captive chimpanzees (42). In humans, many of the traits associated with brain aging display little change until well into old age; cerebral WM, for example, does not start showing noticeable decline until the age of 60 (7, 8). On the other hand, changes in overall brain weight of humans have an earlier onset, between 45 and 50 years of age (29, 31). The great apes analyzed in the current study included specimens within that age range, and a significant proportion was older than 50. Recent evidence indicates that chimpanzees (43) and mountain gorillas (44) can reach advanced postreproductive ages even in the wild, reinforcing the need for comparative studies that sample individuals across the full lifespan to disentangle the contributions of longevity and species-specific aging processes. The current findings support the hypothesis that the evolution of extreme human longevity may amplify the cumulative effects of brain aging, leading to the more pronounced atrophy observed in humans. This effect may be further accentuated in contemporary human populations, where advances in nutrition, public health, and medical care have substantially increased survival into extremely old ages that were likely uncommon throughout much of human evolutionary history. As a result, many individuals now live long enough to experience neurobiological changes that may have had limited opportunity to manifest in ancestral populations or in other great ape species with shorter average lifespans.

In nonhuman primates, age-related microstructural changes in the brain, such as synaptic and dendritic regression, alterations in white matter, and localized cortical thinning are well documented (reviewed in ref. 39), yet these rarely culminate in the substantial, measurable reductions in total brain weight that are typical of aging humans. Our findings reinforce this distinction: great apes showed no evidence of significant age-related brain weight loss. Together, these findings suggest that absolute longevity in humans, combined with species-specific vulnerabilities such as vascular and metabolic diseases, may allow the cumulative effects of microstructural aging to manifest as widespread atrophy and brain weight loss. It remains to be explored how other aspects of somatic aging common to great apes and humans, including cerebral vascular disease, decrease in sensory abilities and mobility (45), relate to the unique patterns of brain aging across species.

Materials and Methods

Subjects and Weight Measurements. Data on brain weight were collected from necropsy records provided by zoos and primate research institutions, as well as weight measurements acquired upon arrival of donated specimens to the National Chimpanzee Brain Resource (www.chimpanzeebrain.org) and Great Ape Neuroscience Project repositories. Only specimens with known age and sex were used in the analysis. If weight for only one hemisphere was available, the values were doubled to get an estimate of the total weight. Although great apes exhibit population-level hemispheric asymmetries, MRI studies indicate that these differences are modest relative to total cerebral volume (46), making hemisphere doubling a reasonable approximation of total brain weight. Part of the chimpanzee sample was previously used in Herndon et al.'s (37) study of brain weight with age in chimpanzees.

The sample consisted of 51 orangutans (*Pongo spp.*), 77 western lowland gorillas (*Gorilla gorilla*), 16 bonobos (*Pan paniscus*), and 457 chimpanzees (*Pan troglodytes*). The captive chimpanzee population in North America is derived predominantly from founders of the Western chimpanzee subspecies (*Pan troglodytes verus*), which contributed approximately 95% of the ancestry of the US captive population (47). Given the low number of specimens in the orangutan sample, the specimens representing Bornean (*Pongo pygmaeus*), Sumatran (*Pongo abelii*), and hybrids were analyzed together. Specimens were included in the analysis regardless of the reported cause of death and neuropathological findings, although we did exclude specimens for which the sample was too physically damaged from necropsy procedures to allow for reliable brain weight measurement. For specimens with comparatively low brain weights, the brains and/or photographic records were inspected to verify that the tissue was grossly intact and that there was no evidence of substantial tissue loss that would compromise the reliability of the measurement. Further details of the sample used in this study are provided in Table 1.

Statistical Analyses. We performed least squares linear regression to determine whether changes in brain weight were associated with age. The data were analyzed separately for each species, using only subjects which reached adult brain size, and separately for females and males. As published reports indicate that adult brain size in gorillas (48) and chimpanzees (49, 50) is reached at the age of 4 y, our analysis included individuals 4 years of age and older. Given that brain growth data for orangutans and bonobos, to the best of our knowledge, have not been published, we relied on the developmental studies on gorillas

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and chimpanzees (48–50), and included orangutans and bonobos older than 4 y. Even though overall brain size in great apes reaches adult-like values at this age, we acknowledge that such individuals remain immature from a life history perspective, with continued somatic, behavioral, and neurobiological development occurring through juvenility and adolescence (51).

Analyses were conducted separately for males and females in gorillas, orangutans, bonobos, and chimpanzees. For each group, we fit ordinary least squares (OLS) linear regressions of brain weight (g) versus age (years). For each model, we report the slope, 95% CI, *P*-values, and coefficient of determination (R^2). Significance was set at $\alpha = 0.05$.

To test whether age-related changes in brain weight exhibited nonlinear patterns, we evaluated quadratic and piecewise alternatives to linear age models. First, we compared linear and quadratic regression models separately for each species and sex. Linear models included age as a single predictor, whereas quadratic models additionally included an age^2 term to capture potential curvature in age-related trajectories. Model support was evaluated using the Akaike Information Criterion (AIC), with differences in AIC (ΔAIC) used to assess relative model fit. When ΔAIC between quadratic and linear models was less than 2, the linear model was preferred for parsimony.

Second, we applied piecewise linear regression allowing a single breakpoint, which was objectively selected using the Bayesian Information Criterion (BIC). Candidate breakpoints were evaluated across the central 80% of the observed age range to minimize edge effects, and piecewise models were only retained if they provided a substantially better fit than a single-slope model (ΔBIC > 2). To reduce the likelihood of overfitting, each segment was required to include a minimum of six individuals. Because our hypothesis specifically concerned late-life decline, only breakpoints associated with a negative slope in the postbreakpoint segment were considered biologically meaningful and interpreted as evidence for age-related reduction in brain weight.

To examine if the variation in brain weight can be explained by differences in body size, we performed least squares regression on individuals 15 y or older, separately for female (n = 147) and male (n = 131) chimpanzees. Next, to assess the effect of body weight on the observed pattern of brain changes with age, we analyzed residual brain weight (from brain ~ body weight) in relation to age. Given that the information on body weight was only available for a subset of our chimpanzee sample, we had to limit our analysis to only one species–chimpanzees–and only to those individuals for which the information on body weight was available. However, these data need to be interpreted with caution, as almost one half of our female chimpanzee sample (n = 50) had body weight above 60 kg, which has been suggested to indicate overweight captive female chimpanzees (52). Whereas comparable data for distinguishing overweight from normal weight are not available for chimpanzee males, 48 out of 94 males in our sample weighed more than 61 kg, which has been reported as the mean weight for captive male chimpanzees (52), suggesting cautious interpretation is warranted.

For a subset of specimens in each species, values were available for both fresh and formalin-fixed brain weight. To examine potential influences of the condition of the tissue (fresh versus formalin-fixed) on our findings, we compiled two datasets for each species, separately for males and females. One dataset consisted of brain weight measurements utilized in the analysis, based on the tissue condition represented by a majority of the sample. For gorillas and orangutans, values for most of the specimens were collected on formalin-fixed specimens; for chimpanzees, measurements for most of the specimens were obtained during necropsy, representing weight of the brain before fixation. In the second dataset, we replaced the values for the specimens for which both fixed and fresh values were available with the opposite to what was included in the first dataset: for gorillas and orangutans, we replaced the values of formalin-fixed specimens with the values for fresh specimens, and for chimpanzees, the values were replaced with those obtained from fixed brains. The only species for which the analysis of the tissue condition was not performed is bonobos, as the values for all except one bonobo specimen were based on formalin-fixed specimens. Average values of each dataset per species are reported in *SI Appendix*, Table S1.

We tested whether tissue condition was confounded with age by comparing the age distributions of specimens for which fresh versus fixed brain weights differed with those of the remaining specimens. Age distributions did not differ significantly in any species-by-sex group (all Mann–Whitney $P > 0.12$). Logistic regression likewise showed that age did not significantly predict whether a specimen had a fresh versus fixed replacement value, either within species while controlling for sex (chimpanzees, $P = 0.071$; gorillas, $P = 0.564$; orangutans, $P = 0.750$) or across the complete sample while controlling for species and sex ($P = 0.118$). Thus, we found no evidence that tissue condition was systematically confounded with age. Next, we performed linear regression to examine change in brain weight with age using both datasets for each species, analyzing females and males separately, and compared the slopes. We did not find a statistically significant difference between the slopes in any of the species (chimpanzee females $P = 0.997$, chimpanzee males $P = 0.753$; gorilla females $P = 0.696$; gorilla males $P = 0.785$; orangutan females $P = 0.979$, orangutan males $P = 0.972$; *SI Appendix*, Table S1). Thus, we concluded that the condition of the tissue did not influence our findings, and the mixture of specimens whose weight was obtained using fresh and formalin-fixed material in each species resulted in similar slopes.

Species and sex differences in slopes were evaluated using ANCOVA contrasts within each species, corrected for multiple comparisons using Tukey HSD. All analyses were performed in Python. Graphs were constructed using GraphPad Prism version 10 for Mac OS (GraphPad Software, Boston, MA).

Data, Materials, and Software Availability. Study data are included in the article and/or supporting information.

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1. A. M. Fjell, K. B. Walhovd, Structural brain changes in aging: Courses, causes and cognitive consequences. *Rev. Neurosci.* **21**, 187–222 (2010).
2. R. Cabeza, N. D. Anderson, J. K. Locantore, A. R. McIntosh, Aging gracefully: Compensatory brain activity in high-performing older adults. *Neuroimage* **17**, 1394–1402 (2002).
3. A. Wingfield, M. Grossman, Language and the aging brain: Patterns of neural compensation revealed by functional brain imaging. *J. Neurophysiol.* **96**, 2830–2839 (2006).
4. P. A. Reuter-Lorenz, K. A. Cappell, Neurocognitive aging and the compensation hypothesis. *Curr. Dir. Psychol. Sci.* **17**, 177–182 (2008).
5. Y. Tang, G. T. Whitman, I. Lopez, R. W. Baloh, Brain volume changes on longitudinal magnetic resonance imaging in normal older people. *J. Neuroimaging* **11**, 393–400 (2001).
6. Y. Taki *et al.*, A longitudinal study of gray matter volume decline with age and modifying factors. *Neurobiol. Aging* **32**, 907–915 (2011).

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7. T. L. Jernigan *et al.*, Effects of age on tissues and regions of the cerebrum and cerebellum. *Neurobiol. Aging* **22**, 581–594 (2001).
8. H. Liu *et al.*, A voxel-based morphometric study of age-and sex-related changes in white matter volume in the normal aging brain. *Neuropsychiatr. Dis. Treat.* **12**, 453–465 (2016).
9. I. Driscoll *et al.*, Longitudinal pattern of regional brain volume change differentiates normal aging from MCI. *Neurology* **72**, 1906–1913 (2009).
10. N. Raz *et al.*, Regional brain changes in aging healthy adults: General trends, individual differences and modifiers. *Cereb. Cortex* **15**, 1676–1689 (2005).
11. N. Raz, P. Ghisletta, K. M. Rodrigue, K. M. Kennedy, U. Lindenberger, Trajectories of brain aging in middle-aged and older adults: Regional and individual differences. *Neuroimage* **51**, 501–511 (2010).
12. B. Jacobs, L. Driscoll, M. Schall, Life-span dendritic and spine changes in areas 10 and 18 of human cortex: A quantitative golgi study. *J. Comp. Neurol.* **386**, 661–680 (1997).
13. B. Jacobs *et al.*, Regional dendritic and spine variation in human cerebral cortex: A quantitative Golgi study. *Cereb. Cortex* **11**, 558–571 (2001).
14. A. Peters, C. Sethares, J. I. Luebke, Synapses are lost during aging in the primate prefrontal cortex. *Neuroscience* **152**, 970–981 (2008).
15. C. C. Sherwood *et al.*, Aging of the cerebral cortex differs between humans and chimpanzees. *Proc. Natl. Acad. Sci. U.S.A.* **108**, 13029–13034 (2011).
16. B. B. Koo *et al.*, Age-related effects on cortical thickness patterns of the rhesus monkey brain. *Neurobiol. Aging* **33**, 200.e23–200.e31 (2012).
17. S. Vickery *et al.*, Chimpanzee brain morphometry utilizing standardized MRI preprocessing and macroanatomical annotations. *Elife* **9**, e60136 (2020).
18. S. Vickery *et al.*, The uniqueness of human vulnerability to brain aging in great ape evolution. *Sci. Adv.* **10**, eado2733 (2024).
19. R. Westerhausen *et al.*, Comparative morphology of the corpus callosum across the adult lifespan in chimpanzees (Pan troglodytes) and humans. *J. Comp. Neurol.* **529**, 1584–1596 (2021).
20. I. August *et al.*, Evolution of human susceptibility to Alzheimer's disease: A review of hypotheses and comparative evidence. *Evol. Anthropol.* **34**, e22054 (2025).
21. P. R. Hof, J. H. Morrison, Neocortical neuronal subpopulations labeled by a monoclonal antibody to calbindin exhibit differential vulnerability in Alzheimer's disease. *Exp. Neurol.* **111**, 293–301 (1991).
22. J. H. Morrison *et al.*, A monoclonal antibody to non-phosphorylated neurofilament protein marks the vulnerable cortical neurons in Alzheimer's disease. *Brain Res.* **416**, 331–336 (1987).
23. M. J. West, P. D. Coleman, D. G. Flood, J. C. Troncoso, Differences in the pattern of hippocampal neuronal loss in normal ageing and Alzheimer's disease. *Lancet* **344**, 769–772 (1994).
24. A. M. Hedman *et al.*, Human brain changes across the life span: A review of 56 longitudinal magnetic resonance imaging studies. *Hum. Brain Mapp.* **33**, 1987–2002 (2012).
25. S. M. Resnick *et al.*, Longitudinal magnetic resonance imaging studies of older adults: A shrinking brain. *J. Neurosci.* **23**, 3295–3301 (2003).
26. A. Irimia *et al.*, The indigenous South American Tsimane exhibit relatively modest decrease in brain volume with age despite high systemic inflammation. *J. Gerontol. A Biol. Sci. Med. Sci.* **76**, 2147–2155 (2021).
27. P. Hartmann *et al.*, Das Normgewicht des Gehirns beim Erwachsenen in Abhängigkeit von Alter, Geschlecht. *Körpergrösse und Gewicht. Der Pathologie* **15**, 165–170 (1994).
28. K. C. Ho, U. Roessmann, J. V. Straumfjord, G. Monroe, Analysis of brain weight. I. Adult brain weight in relation to sex, race, and age. *Arch. Path. Lab. Med.* **104**, 635–639 (1980).
29. A. S. Dekaban, D. Sadowsky, Changes in brain weights during the span of human life: Relation of brain weights to body heights and body weights. *Ann. Neurol.* **4**, 345–356 (1978).
30. F. W. Appel, E. M. Appel, Intracranial variation in the weight of the human brain. *Hum. Biol.* **14**, 48–68 (1942).
31. H. Pakkenberg, J. Voigt, Brain weight of the Danes: A forensic material. *Acta Anat.* **56**, 297–307 (1964).
32. J. G. Herndon, J. Tigges, S. A. Klumpp, D. C. Anderson, Brain weight does not decrease with age in adult rhesus monkeys. *Neurobiol. Aging* **19**, 267–272 (1998).
33. K. E. Langergraber *et al.*, Generation times in wild chimpanzees and gorillas suggest earlier divergence times in great ape and human evolution. *Proc. Natl. Acad. Sci. U.S.A.* **109**, 15716–15721 (2012).
34. M. E. Steiper, N. M. Young, Primate molecular divergence dates. *Mol. Phylogenet. Evol.* **41**, 384–394 (2006).
35. B. Bogin, Childhood, adolescence, and longevity: A multilevel model of the evolution of reserve capacity in human life history. *Am. J. Hum. Biol.* **21**, 567–577 (2009).
36. H. Kaplan, "The evolution of the human life course" in *Between Zeus and the Salmon: The Biodemography of Longevity*, K. W. Wachter, C. E. Finch, Eds. (National Academy Press, 1997), pp. 175–211.
37. J. G. Herndon, J. Tigges, D. C. Anderson, S. A. Klumpp, H. M. McClure, Brain weight throughout the life span of the chimpanzee. *J. Comp. Neurol.* **409**, 567–572 (1999).
38. R. L. Holloway, "Evolution of the human brain" in *Handbook of Human Symbolic Evolution*, A. Lock, C. R. Peters, Eds. (Oxford University Press, 1996), pp. 74–114.
39. C. Freire-Cobo *et al.*, Comparative neuropathology in aging primates: A perspective. *Am. J. Primatol.* **83**, e23299 (2021).
40. M. M. Autrey *et al.*, Age-related effects in the neocortical organization of chimpanzees: Gray and white matter volume, cortical thickness, and gyrification. *Neuroimage* **101**, 59–67 (2014).
41. K. Hill *et al.*, Mortality rates among wild chimpanzees. *J. Hum. Evol.* **40**, 437–450 (2001).
42. K. Havercamp *et al.*, Longevity and mortality of captive chimpanzees in Japan from 1921 to 2018. *Primates* **60**, 525–535 (2019).
43. B. M. Wood *et al.*, Demographic and hormonal evidence for menopause in wild chimpanzees. *Science* **382**, eadd5473 (2023).
44. N. Smit, M. M. Robbins, Post-reproductive lifespan in wild mountain gorillas. *Proc. Natl. Acad. Sci. U.S.A.* **122**, e25109981 (2025).
45. L. J. Lowenstine, R. McManamon, K. A. Terio, Comparative pathology of aging great apes: Bonobos, chimpanzees, gorillas, and orangutans. *Vet. Pathol.* **53**, 250–276 (2016).
46. W. D. Hopkins, A. Meguerditchian, Handedness and brain asymmetries in nonhuman primates. *Handb. Clin. Neurol.* **208**, 197–210 (2025).
47. J. J. Ely *et al.*, Subspecies composition and founder contribution of the captive US chimpanzee (Pan troglodytes) population. *Am. J. Primatol.* **67**, 223–241 (2005).
48. S. C. McFarlin *et al.*, Early brain growth cessation in wild Virunga mountain gorillas (Gorilla beringei beringei). *Am. J. Primatol.* **75**, 450–463 (2013).
49. S. R. Leigh, Brain growth, life history, and cognition in primate and human evolution. *Am. J. Primatol.* **62**, 139–164 (2004).
50. S. Neubauer *et al.*, Brief communication: Endocranial volumes in an ontogenetic sample of chimpanzees from the Taï Forest National Park, Ivory Coast. *Am. J. Phys. Anthropol.* **147**, 319–325 (2012).
51. M. E. Thompson, K. Sabbi, "19. Evolutionary demography of the great apes" in *Human Evolutionary Demography*, O. Burger, R. Lee, R. Sear, Eds. (Open Book Publishers, 2024), pp. 423–474.
52. E. N. Videan, J. Fritz, J. Murphy, Development of guidelines for assessing obesity in captive chimpanzees (Pan troglodytes). *Zoo Biol.* **26**, 93–104 (2007).

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